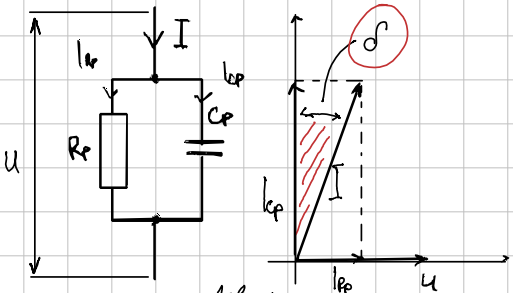
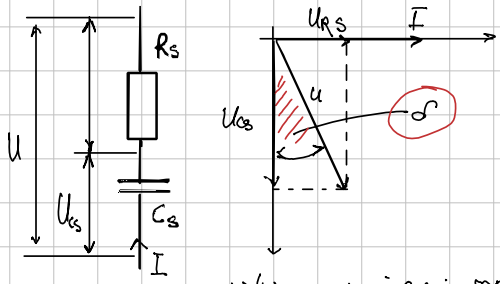
Mjerni kondenzatori

Labos 6.
Određivanje kuta gubitaka integracijskom metodom

→ Nadomjesna shema realnog kondenzatora



paralelni



serijski nadomjesni spoj

→ Gubitke na kondenzatoru definiramo tangensom kuta gubitaka tgδ

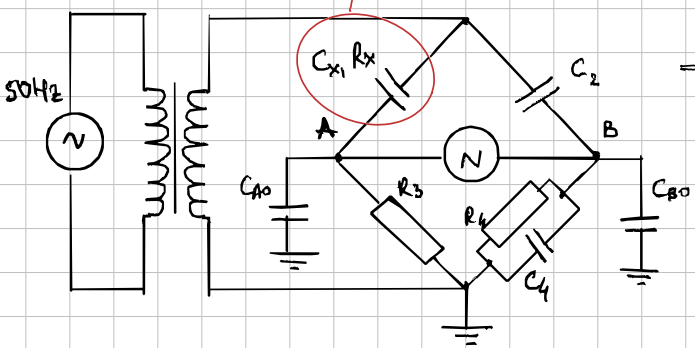
$$tg\delta = \frac{I_{RP}}{I_{CP}} = \frac{U}{R_P} \cdot \frac{1}{U\omega C_P} = \underline{\underline{\frac{1}{\omega R_P C_P}}}$$

$$tg\delta = \frac{U_{RS}}{U_{CS}} = \frac{I \cdot R_S}{\frac{I}{\omega \cdot C_S}} = \underline{\underline{\omega R_S C_S}}$$

Scheringov most

mjereni kondenzator nadomješten serijskim spojem C_x i R_x

C_x želimo izmjeriti



⇒ radi se za mjerenje kuta gubitaka el. opreme i izolacijskih materijala, naročito pri visokim naponima i pri freker mreže od 50 Hz

• neovisno ugađanje postiže se s R_4 i C_4

$$Z_1 = R_x + \frac{1}{j\omega C_x} \qquad Z_2 = \frac{1}{j\omega C_2}$$

$$Z_3 = R_3 \qquad Z_4 = R_4 \| C_4 = \frac{R_4}{1 + j\omega C_4 R_4}$$

U ravnoteži vrijedi:

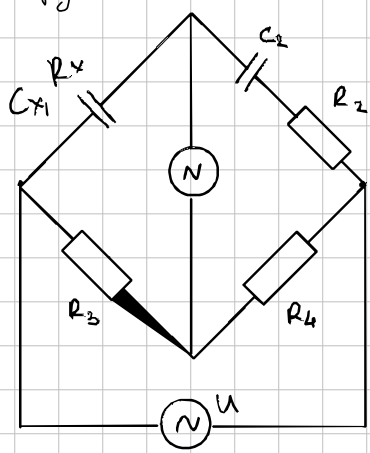
$$R_x = R_3 \frac{C_4}{C_2} \qquad C_x = C_2 \frac{R_4}{R_3}$$

$$tg\delta = \omega R_x C_x = \omega R_4 C_4$$

Wienov most

- mjereni kondenzator nadomješten serijskim spojem C_x i R_x (gubitaka)

- neovisno ugađanje postiže se s R_2 i C_2



→ Impedancije u granama mosta:

$$Z_1 = R_x + \frac{1}{j\omega C_x} \qquad Z_2 = R_2 + \frac{1}{j\omega C_2}$$

$$Z_3 = R_3 \qquad Z_4 = R_4$$

$$R_x = \frac{R_2 R_3}{R_4} \qquad C_x = C_2 \frac{R_4}{R_3}$$

$$\operatorname{tg}\delta = \omega R_x C_x = \omega R_2 C_2$$

$\delta \ll$; dobro ugađanje se postiže i s promjenom R_2 i R_3 ili R_2 i R_4

Neuravnoteženi mostovi, njihove osjetljivosti